MINI50: informal training only. FOX10: uncalibrated Foxoring candidate.
UNBUILT / 35 x 35 mm / 2-layer 1.6 mm / independent fourth family.
No normal-power finish beacon. See README and first-article procedure.

01 Battery, reverse polarity and switched RF supply

File: 01_power.kicad_sch

02 Autonomous control and pogo fixture

File: 02_control.kicad_sch

03 Programmable source, eight isolated RF outputs and envelope

File: 03_rf.kicad_sch

04 Seven-pole LPF and selectable antenna network

File: 04_output.kicad_sch

**ARDF-TX**

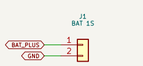
Sheet: /
File: Mini80.kicad_sch

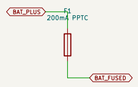
***Title: Mini80 autonomous low-power 80m transmitter***

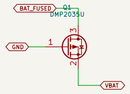
| Size: A3 | Date: 2026-10-21 | **Rev: M80-E1** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 1/5 |

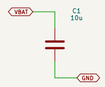
# 01 Battery, reverse polarity and switched RF supply

Protected 1S cell only. 3.2–4.2 V useful range. No charger. Q1 reverse protection.
BOOST_EN and CLK_EN have hardware pull-downs. TPS61099 has true shutdown disconnect.
MCP1700 3.0 V always-on logic: 4.9/5.05 V boost only during prepared slots.



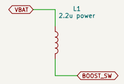


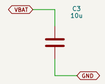


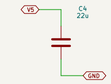


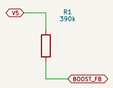


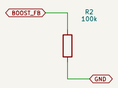


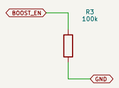


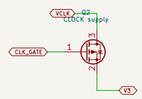



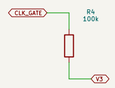



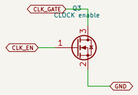



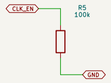



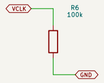



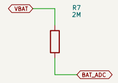



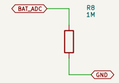



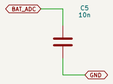




M80-E1: REVIEW / ALL hardware measurements NOT RUN / unbuilt engineering candidate.

Evaluation review only; unbuilt engineering candidate

**ARDF-TX**

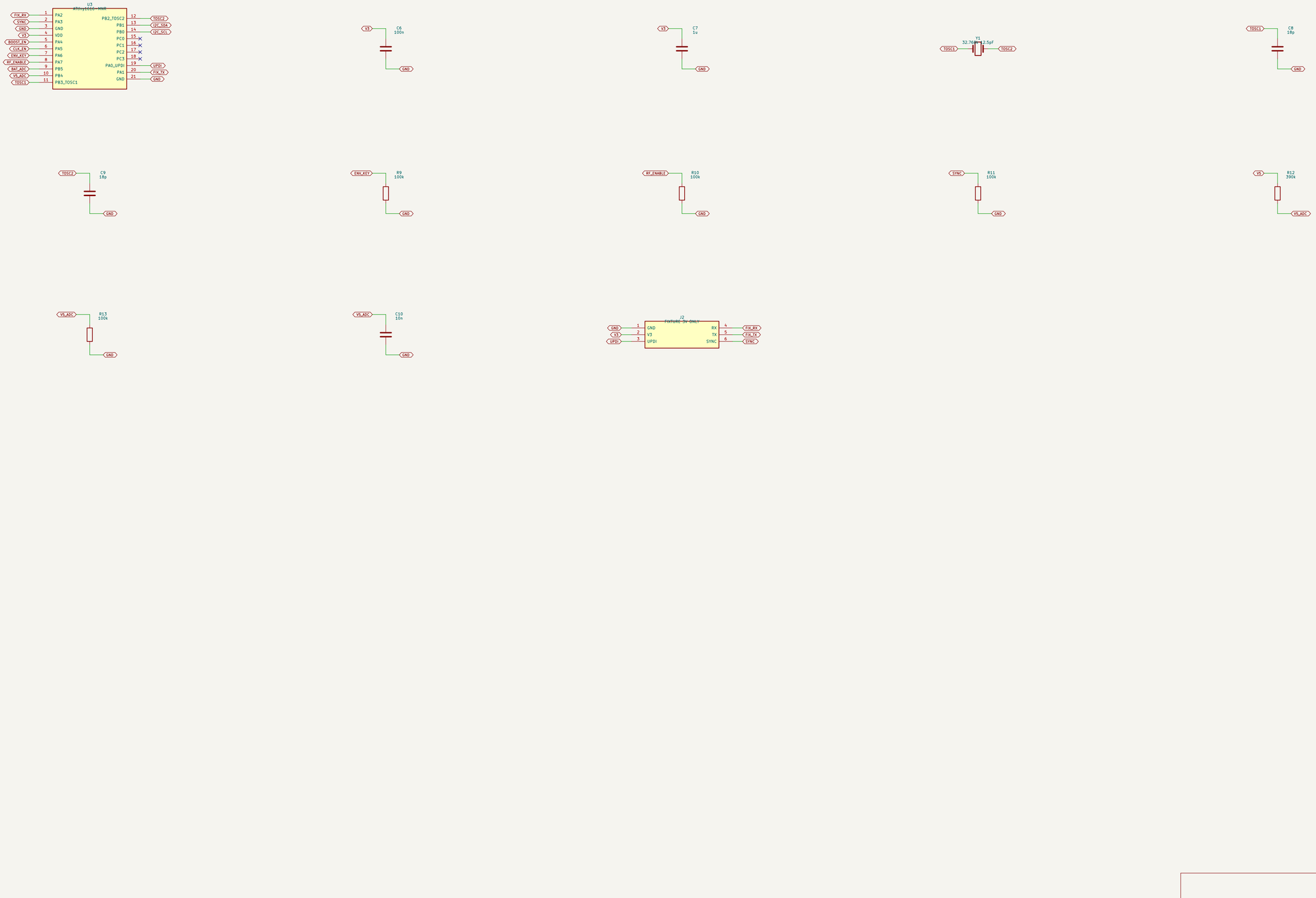
Sheet: /01 Battery, reverse polarity and switched RF supply/
File: 01_power.kicad_sch

***Title: 01 Battery, reverse polarity and switched RF supply***

| Size: A2 | Date: 2026-10-21 | **Rev: M80-E1** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 2/5 |

# 02 Autonomous control and pogo fixture

ATtiny1616-MNR; VQFN20 3x3, 0.4mm pitch, exposed pad grounded. UPDI remains UPDI.
32.768 kHz crystal provides RTC schedule. UART uses alternate PA1/PA2 mapping.
One image: CRC configuration in EEPROM; reset always requires fresh fixture SYNC/ARM.




M80-E1: REVIEW / ALL hardware measurements NOT RUN / unbuilt engineering candidate.

Evaluation review only; unbuilt engineering candidate

**ARDF-TX**

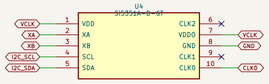
Sheet: /02 Autonomous control and pogo fixture/
File: 02_control.kicad_sch

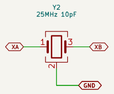
***Title: 02 Autonomous control and pogo fixture***

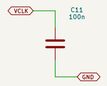
| Size: A2 | Date: 2026-10-21 | **Rev: M80-E1** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 3/5 |

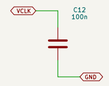
# 03 Programmable source, eight isolated RF outputs and envelope

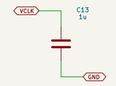
SI5351A-B-GT factory outputs disabled. CLK0 is the only used output: HCT TTL VIH >=2V.
Eight outputs each have a separate 100R ballast (MINI50). 560R (FOX10 initial).
Q4 RC supply envelope is experimental: verify HCT behavior below 4.5V during edges.
RF_ENABLE hardware pull-down -> Q6 OFF -> both OE_N HIGH. Oscillator retune only with RF inhibited.

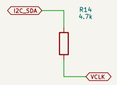


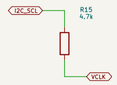



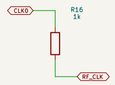



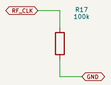



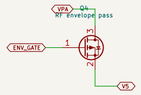



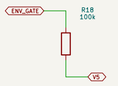



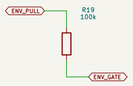



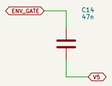



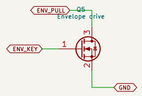



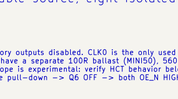











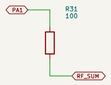


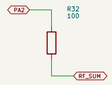


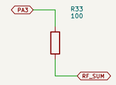


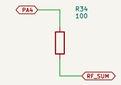


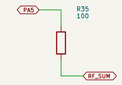


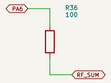



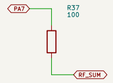



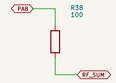




M80-E1: REVIEW / ALL hardware measurements NOT RUN / unbuilt engineering candidate.

Evaluation review only: unbuilt engineering candidate

**ARDF-TX**

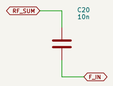
Sheet: /03 Programmable source, eight isolated RF outputs and envelope/
File: 03_rf.kicad_sch

***Title: 03 Programmable source, eight isolated RF outputs and envelope***

| Size: A2 | Date: 2026-10-21 | **Rev: M80-E1** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 4/5 |

# 04 Seven-pole LPF and selectable antenna network

50-ohm reference port TP_RF precedes antenna tuning. No simultaneous dummy load and antenna.
LPF: 1u/1200p/3.3u/1000p/3.3u/820p/1u. C0G RF capacitors.
MINI: optional 2.2u/470p L-match, then 33u + optional 10u/4.7u/15u loading.
FOX: bypass loading, select coupling capacitor separately; start 30cm vertical, field-calibrate.
Default R50/51/52/53/54 fitted bypasses for 50-ohm first-article tests. Loading coils DNP initially.















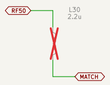


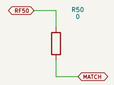






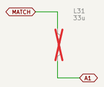


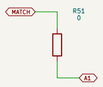




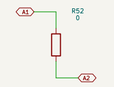






























































M80-E1: REVIEW / ALL hardware measurements NOT RUN / unbuilt engineering candidate.

Evaluation review only: unbuilt engineering candidate

**ARDF-TX**

Sheet: /04 Seven-pole LPF and selectable antenna network/
File: 04_output.kicad_sch

***Title: 04 Seven-pole LPF and selectable antenna network***

| Size: A2 | Date: 2026-10-21 | **Rev: M80-E1** |
|---|---|---|
| KiCad E.D.A. 10.0.6 | | Id: 5/5 |